



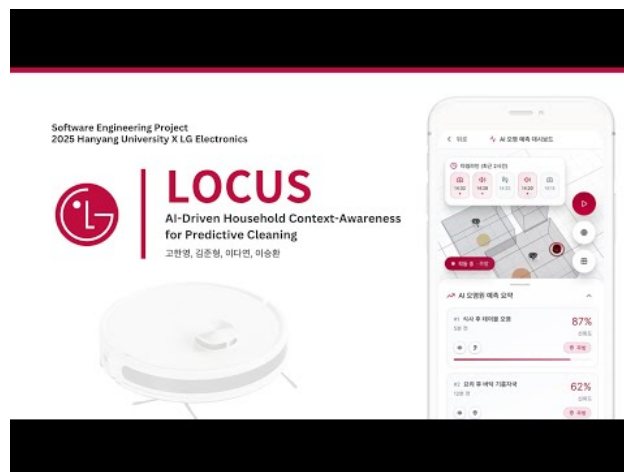
LOCUS

| AI-Driven Household Context-Awareness for Predictive Cleaning

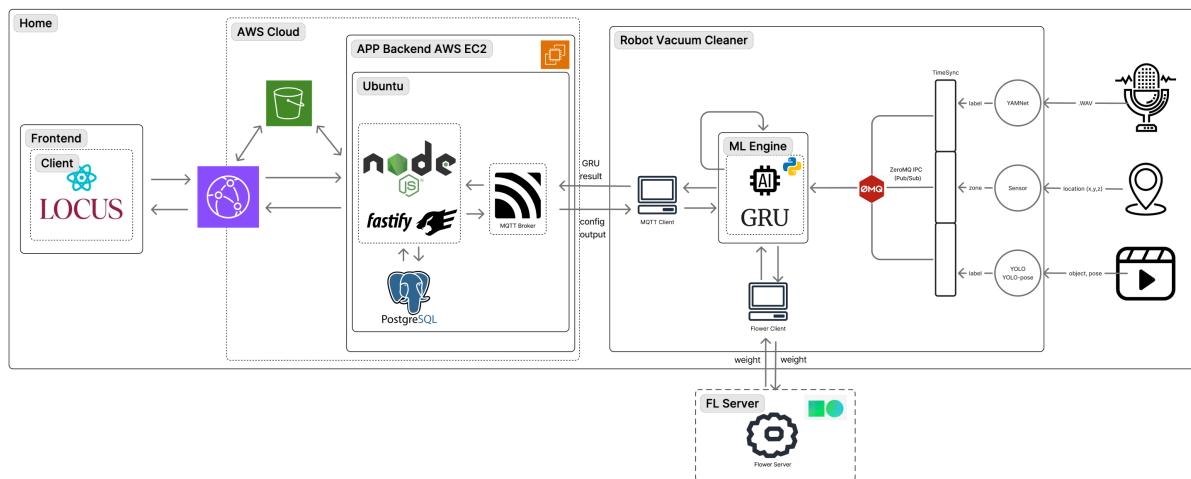
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Video Presentation



Video Link: <https://www.youtube.com/watch?v=xoD5vsCmPoc>



I. Introduction

Motivation: Why are we doing this?

This is to address current limitations in robotic vacuum cleaners, which have evolved from simple tools to essential smart appliances but remain reactive rather than proactive. Despite hardware advances, current commercial models only operate on preset schedules or direct commands, lacking the ability to respond immediately to dynamic contamination scenarios or understand context such as when, where, and why dirt occurs. Furthermore, privacy concerns arise because conventional systems transmit sensitive data to cloud servers. This approach enables proactive, context-aware cleaning and ensures data privacy by processing sensitive information locally.

What do we want to see at the end?

- Predictive Cleaning Based on Multi-modal Context: Complete an 'Active Partner' system that goes beyond simple scheduling by fusing visual(YOLO), auditory(YAMNet), and positional(Apple ArKit) information to predict the timing and area of contamination and respond accordingly.
- On-device AI-based Privacy Assurance: Process sensitive raw data (video, audio) internally on the edge device without transmitting it to the cloud, and fundamentally prevent privacy leakage by sharing only non-personally identifiable model weights through Federated Learning.
- User-specific Environmental Adaptation: Learn the semantic labels assigned to spaces and the user's living patterns via an app, providing a personalized cleaning solution optimized for each household.

II. Datasets

1. YOLO Datasets (Visual Feature Extraction)

- Objective: Household object recognition and contamination detection for the visual perception of robotic vacuums.
- Final Data: HomeObjects-3K + HD10K

Dataset	Purpose	Total	Train / Val	Source
HomeObjects-3K	Household Object Recognition	2,689	2,285 / 404	Ultralytics, License: AGPL-3.0
HD10K	Contamination Detection	6,000	4,800 / 1,200	IROS 2022 Dataset
Final Data	YOLO Training	8,689	7,085 / 1,604	

- Classes(14): bed, sofa, chair, table, lamp, tv, laptop, wardrobe, window, door, potted_plant, photo_frame, solid_waste, liquid_stain
- YOLO Training Settings:
 - Base Model: YOLOv11n (COCO pretrained)

- Freeze: 10 layers
- Learning Rate: 0.001
- Epochs: 50 (Early Stopping → 24)
- Image Size: 640×640, Batch Size: 4

2. YAMNet Datasets (Auditory Feature Extraction)

- Objective: Context awareness based on household environmental sounds for robotic vacuums (Auditory).
- Source: Google AudioSet → Refined Home Audio Dataset constructed after background/noise removal.
- Final Data: Total 22,212 clips (Train 17,770 / Val 2,221 / Test 2,221)
 - Classes(17): speech, door, dishes, cutlery, chopping, frying, microwave, blender, water_tap, sink, toilet_flush, telephone, chewing, television, footsteps, vacuum, hair_dryer
 - Input Shape: (N, 1024)
- YAMNet Training Settings:
 - Backbone: YAMNet (MobileNetV1-based, 1024-dim, Frozen)
 - Head: Dense(512-ReLU) → Dropout(0.2) → Dense(17-Sigmoid)
 - Loss: Binary Cross Entropy (Weighted for class imbalance)
 - Epochs: 100 (Early Stopping)
 - Final Model: TFLite FP16, 17MB

3. GRU Datasets (Context-Based Cleaning Prediction)

- Objective: Prediction of zones requiring cleaning based on context.
- Input: YOLO(14) + YAMNet(17) + Pose(51) + GPS(4) + Time(10) → 160-dimensional context vector generated by AttentionContextEncoder.
- Sequence Length: 30 (Each timestep = 1 context vector)
- Output: Probability of cleaning necessity for 4 zones (balcony, bedroom, kitchen, living_room), Multi-label binary classification.
- Data Count: Train 1,120 / Val 280

III. Methodology

1. Choice of Algorithms & Models

1. Visual Context: YOLOv11n-pose (Quantized)

- We selected YOLOv11n, the most lightweight version of the YOLO model family. Specifically, we adopted the Pose version to go beyond simple object detection and utilize Keypoints to precisely analyze human actions (e.g., 'sitting', 'standing').
- Optimization: We applied TensorFlow Lite Quantization for deployment on edge device environments. This effectively reduced the model size while maintaining inference speed.

2. Audio Context (Transfer learning, Quantized)

- YAMNet is designed based on the MobileNetV1 architecture, which is optimized for mobile devices. It offers the advantage of efficiently classifying household environmental sounds (such as dishwashing) without the high computational costs associated with large audio transformer models.

3. Sequential Prediction: GRU (Gated Recurrent Unit)

- We selected GRU over LSTM for time-series prediction. GRU has a simpler gate structure than LSTM, requiring less computation, making it more suitable for edge devices. Simultaneously, it effectively captures the Temporal Dependencies of household living patterns.

4. Data Synchronization: TimeSyncBuffer (Custom Algorithm)

- Standard synchronization is difficult because vision sensors (~30fps) and audio sensors (~1s) operate at different frequencies. Therefore, we implemented a custom Sliding Window algorithm that aligns asynchronous data

streams based on timestamps (using the LOCF strategy) to merge them into a single context vector.

5. Federated Learning: FedPer Architecture

- We adopted the FedPer architecture to balance privacy protection and model performance. Unlike standard FedAvg, this approach separates the model into a Base Layer (Shared) and a Head Layer (Personal), allowing the robot to learn unique household habits without sharing sensitive label distribution information.

IV. Evaluation & Analysis

1. YOLO Object Detection Model Results

- Validation Metrics

Metric	Score
Precision	0.637
Recall	0.544
mAP50	0.566
mAP50-95	0.389

- Loss Values

Loss	Values
Box Loss	0.988
Class Loss	0.793
DFL Loss	0.954

⇒ Visual object recognition performance is at a moderate level.

2. YAMNet Model Results

- Overall Performance Summary

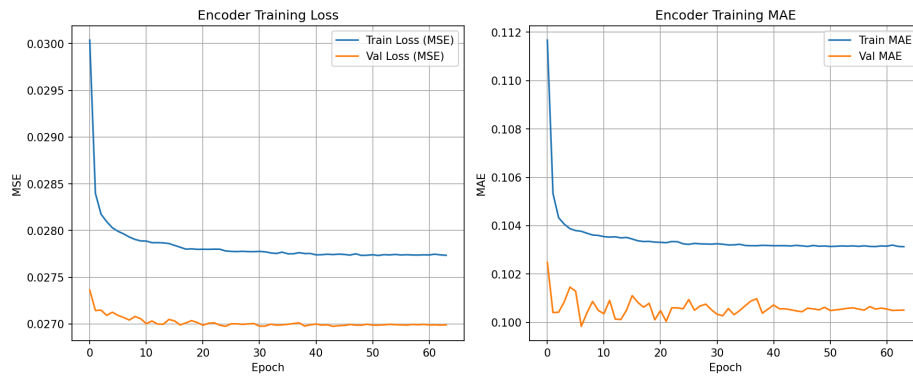
Item	Value
Total Samples	22,212
Macro AUC	0.9899
Macro AUPR	0.9705

- Per-Class Performance(Top5)

Rank	Class	AUC	AUPR
1	vacuum	0.9980	0.9957
2	hair_dryer	0.9981	0.9941
3	toilet_flush	0.9962	0.9911
4	blender	0.9948	0.9886
5	telephone	0.9954	0.9870

⇒ Accurately classifies auditory events.

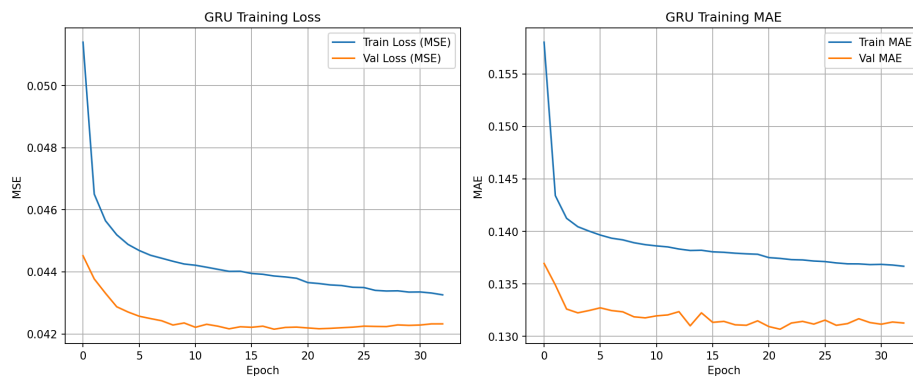
3. AttentionContextEncoder Model Results



- Training Epochs: 64 (Best Epoch: 44)
- Loss Convergence: Train 0.0300 $\rightarrow$ 0.0277, Val 0.0274 $\rightarrow$ 0.0270
- MAE Performance: Train 0.1117 $\rightarrow$ 0.1031, Val 0.1025 $\rightarrow$ 0.1005
- Best Performance: Val Loss 0.0270, Val MAE 0.1005 (Epoch 44)
- Stable convergence with minimal overfitting gap

$\Rightarrow$ The AttentionContextEncoder effectively fuses multi-modal sensor data (YOLO, YAMNet, Pose, GPS, Time) into 160-dimensional context vectors with low prediction error.

4. GRU Model Results



- Training Epochs: 33 (Best Epoch: 18)
- Loss Convergence: Train 0.0514 $\rightarrow$ 0.0433, Val 0.0445 $\rightarrow$ 0.0423
- MAE Performance: Train 0.1580 $\rightarrow$ 0.1367, Val 0.1369 $\rightarrow$ 0.1313
- Best Performance: Val Loss 0.0422, Val MAE 0.1311 (Epoch 18)
- Consistent train-val loss gap indicates good generalization

$\Rightarrow$ The GRU-based sequential model successfully predicts zone-specific pollution probabilities from 30-timestep context sequences, achieving Mean Absolute Error of 0.131 on validation data.

V. Related Work

This project builds upon several foundational AI technologies and research advances in computer vision, audio processing, federated learning, and multi-modal sensor fusion for robotics applications.

1. Object Detection and Pose Estimation

YOLOv11 for Robotic Vision: YOLOv11, developed by Ultralytics and unveiled at YOLO Vision 2024 conference, represents the state-of-the-art in real-time object detection. Our project leverages YOLOv11n-pose, which supports 17-keypoint human pose estimation trained on the COCO keypoints dataset. YOLOv11 achieves 25-40% latency reduction compared to YOLOv10 while maintaining higher accuracy with 22% fewer parameters, making it ideal for edge deployment in robotic

applications. The pose estimation capability enables robots to interpret physical actions in real-time, which is critical for understanding household activities and predicting contamination patterns.

Deployment Optimization: Ultralytics provides comprehensive documentation for deploying YOLO models on edge devices through TensorFlow Lite quantization, enabling real-time inference on resource-constrained platforms such as robot vacuums and mobile devices.

2. Audio Event Classification

YAMNet and AudioSet: YAMNet is a pretrained deep neural network developed by Google that predicts 521 audio event classes based on the AudioSet-YouTube corpus. Built on the MobileNet_v1 depthwise-separable convolution architecture, YAMNet employs 3.7M weights and performs 69.2M multiplies for each 960ms input frame at 16kHz sampling rate. The model was trained on 1,574,587 ten-second YouTube excerpts from AudioSet's unbalanced train segments, covering diverse environmental sounds including household activities.

Transfer Learning Capabilities: TensorFlow Hub provides YAMNet as a reusable module for transfer learning on custom audio classification tasks. Our project fine-tunes YAMNet on household-specific sounds (17 classes including cooking, dishwashing, vacuum operation) to achieve domain-specific performance superior to the generic pretrained model.

3. Federated Learning with Personalization

FedPer Architecture: Arivazhagan et al. (2019) introduced Federated Learning with Personalization Layers (FedPer) in their seminal work "Federated Learning with Personalization Layers" (arXiv:1912.00818). FedPer addresses statistical heterogeneity in federated learning by separating the model into a shared base layer (feature extractor trained collaboratively across clients) and personalized head layers (trained locally and never shared). This architecture enables each client to adapt to local data distributions while benefiting from collective learning.

Privacy-Preserving Personalization: Recent extensions of FedPer, such as FedPerGNN for graph neural networks, demonstrate 4.0-9.6% lower prediction errors compared to state-of-the-art privacy-preserving methods under satisfactory privacy budgets. The base + personalization layer paradigm effectively combats the adverse effects of statistical heterogeneity in federated networks, which is critical for household-specific cleaning pattern learning in our system.

4. Recurrent Neural Networks for Time Series Prediction

GRU for Smart Home Applications: Gated Recurrent Units (GRUs) have become the preferred architecture for time-series forecasting in smart home and smart grid contexts due to their simpler structure compared to LSTMs, easier training process, and faster operation rates while maintaining competitive performance. GRUs effectively retain long-term dependencies in sequences crucial for predicting future values, making them ideal for sequential pattern learning.

Hybrid Approaches: Recent research demonstrates that hybrid CNN-GRU frameworks outperform conventional LSTM and GRU neural networks for building load prediction, with CNNs extracting useful information from high-uncertainty data and GRUs providing superior time-series forecasting. Studies on energy consumption prediction show that bidirectional GRU approaches and attention mechanisms allow models to focus on the most important time steps, improving the ability to capture complex temporal patterns in household behavior.

5. Multi-Modal Sensor Fusion and Edge AI

Recent Advances in 2024: State-of-the-art multi-modal sensor fusion systems demonstrate remarkable performance improvements through edge computing integration. Recent work shows multimodal fusion algorithms achieving 98.7% sorting accuracy with distributed edge computing architectures enabling 3.2ms average response time through local processing optimization. Edge computing integration with multi-modal perception enables unprecedented capabilities in real-time adaptation and decision-making, achieving sub-20ms response times for complex fusion operations.

Sensor Fusion Architectures: Research in robotic vision employs various fusion strategies including encoder-decoder frameworks, attention mechanisms, and graph neural networks to process heterogeneous sensory inputs (RGB, Depth, LiDAR, GPS, IMU). These multimodal fusion approaches support core robotic vision tasks such as 3D semantic scene understanding, SLAM, object detection, navigation, and manipulation.

Tactile-Visual Integration: Cutting-edge research demonstrates multimodal tactile sensors utilizing flexible thin-film thermistors for pressure, temperature, texture, and slip detection, achieving ultrasensitive (0.05 mm/s) and ultrafast (4ms) slip sensing for dexterous grasping control. This highlights the trend toward diverse sensor modalities beyond vision and audio.

Market Growth: The multi-modal sensor fusion market is projected to grow from USD 7.1 billion in 2024 to USD 42.5 billion by 2033, reflecting the expanding integration of these technologies across industrial sectors including robotics, autonomous vehicles, healthcare, and smart home systems.

6. Datasets and Benchmarks

IROS 2022 HD10K Dataset: Our visual pollution detection leverages the IROS 2022 dataset specifically designed for household dirt detection, providing 6,000 annotated images of solid waste and liquid stains in residential environments. This domain-specific dataset significantly improves detection accuracy compared to general-purpose object detection datasets.

Ultralytics HomeObjects-3K: This dataset provides 2,689 household object images covering common furniture and appliances, enabling the system to understand room context and spatial semantics beyond simple pollution detection.

Google AudioSet: AudioSet remains the largest-scale collection of manually-annotated audio events, containing over 2 million human-labeled 10-second YouTube clips spanning 632 audio event classes. This dataset forms the foundation for YAMNet's pretraining and enables effective transfer learning to household-specific audio classification.

Connection to Our Work

Our system uniquely combines these established technologies into an integrated on-device AI pipeline:

- **YOLOv11n-pose** provides visual context understanding through object detection and human activity recognition
- **YAMNet** offers auditory context awareness by classifying household environmental sounds
- **FedPer** enables privacy-preserving personalization without transmitting sensitive household data
- **GRU** learns temporal dependencies between multi-modal contexts and zone-specific pollution patterns
- **Multi-modal fusion** at the edge ensures real-time responsiveness without cloud dependency

The novelty lies not in individual components but in their integration into a cohesive, privacy-preserving, context-aware predictive system that transforms reactive robotic cleaning into proactive household automation.

VI. Conclusion

Summary of Achievements

This project successfully developed an AI-driven predictive cleaning system that transforms robotic vacuum cleaners from reactive tools into proactive household partners. By integrating multi-modal sensor fusion (YOLOv11n-pose for visual context, YAMNet for auditory context, and GPS/IMU for spatial context), the system achieves context-aware pollution prediction rather than relying solely on fixed schedules.

The key technical achievements include:

1. **Multi-modal Context Fusion:** The AttentionContextEncoder successfully combines heterogeneous sensor data (visual, audio, pose, location, temporal) into unified 160-dimensional context vectors with a validation MAE of 0.1005, demonstrating effective cross-modal learning.
2. **Sequential Pollution Prediction:** The GRU-based model learns temporal patterns from 30-timestep sequences and predicts zone-specific cleaning needs with a validation MAE of 0.1313, enabling the system to anticipate contamination before it becomes severe.
3. **Privacy-Preserving Personalization:** By implementing Federated Personalization (FedPer) architecture, the system ensures that sensitive household data never leaves edge devices while still enabling personalized learning through on-device Head Layer fine-tuning.
4. **Real-time Processing Pipeline:** The TimeSyncBuffer algorithm successfully synchronizes asynchronous sensor streams (visual at ~30fps, audio at ~1s intervals) within ± 500 ms tolerance, enabling consistent real-time inference.

Performance Evaluation

The evaluation results demonstrate that our AI models achieve practical performance levels:

- **YAMNet Audio Classification:** Macro AUC of 0.9899 indicates excellent capability in recognizing household sounds (cooking, dishwashing, vacuum operation, etc.), providing reliable auditory context.
- **YOLO Visual Detection:** mAP50 of 0.566 achieves moderate performance in detecting household objects and pollution (solid waste, liquid stains), sufficient for context-aware decision making.
- **Context Encoder & GRU Prediction:** Combined MAE of approximately 0.10-0.13 demonstrates that the system can reasonably estimate pollution probability, though there is room for improvement in prediction accuracy.

System Significance

This research demonstrates three important contributions to the field of intelligent home robotics:

1. **Privacy by Design:** Unlike cloud-based systems, all AI inference occurs on edge devices (robot vacuum and smartphone tracker), eliminating the transmission of sensitive camera and microphone data to external servers.
2. **Adaptive Personalization:** The FedPer architecture enables household-specific customization without compromising privacy, allowing the robot to learn unique living patterns while sharing only non-identifiable model weights during federated learning.
3. **Proactive Automation:** By predicting when and where cleaning is needed based on multi-modal context, the system reduces manual intervention and improves user experience through intelligent automation.

Limitations

Despite the achievements, several limitations should be acknowledged:

1. **Simulation Environment:** The current implementation uses Webots simulation rather than real-world robot deployment, which may not fully capture the complexity and variability of actual household environments.
2. **Dataset Scale:** The training datasets (1,120 training sequences for GRU, 8,689 images for YOLO) are relatively small compared to large-scale commercial systems, potentially limiting model generalization.
3. **Prediction Accuracy:** The GRU model's MAE of 0.131 indicates approximately 13% average error in pollution probability prediction, suggesting that the system may occasionally miss cleaning opportunities or trigger unnecessary cleaning.
4. **Computational Constraints:** While optimized for edge devices, running multiple AI models simultaneously (YOLO, YAMNet, GRU) requires significant computational resources, which may challenge deployment on low-power robot hardware.
5. **Limited Sensor Modalities:** The current system does not incorporate chemical sensors (odor detection) or advanced environmental sensors (humidity, temperature) that could further improve context awareness.

Future Work

Several directions for future research and development include:

1. **Real-world Deployment:** Implementing the system on actual robot vacuum hardware to validate performance under real household conditions, including varying lighting, acoustic noise, and floor types.
2. **Model Improvement:** Exploring advanced architectures such as Transformer-based models for better sequence modeling, and investigating larger-scale datasets to improve prediction accuracy.
3. **Additional Sensor Integration:** Incorporating chemical sensors for odor detection, depth cameras for 3D environment understanding, and contact sensors for physical contamination verification.
4. **Enhanced Federated Learning:** Implementing differential privacy mechanisms to provide mathematical privacy guarantees, and exploring more sophisticated aggregation strategies beyond FedPer.
5. **User Interaction:** Developing intuitive feedback mechanisms that allow users to correct prediction errors, enabling continuous improvement through human-in-the-loop learning.
6. **Energy Optimization:** Implementing model quantization and adaptive inference strategies to reduce power consumption, enabling longer battery life for autonomous operation.
7. **Cross-household Generalization:** Investigating transfer learning techniques to improve initial performance in new households by leveraging patterns learned from other homes while preserving privacy.

This project establishes a foundation for privacy-preserving, context-aware, and personalized robotic cleaning systems. While current performance demonstrates proof-of-concept feasibility, continued research in model accuracy, real-world deployment, and user interaction will be essential for practical commercial adoption.
